The Hong Kong University of Science and Technology
IEDA 1250, Summer 2026
Midterm Exam

Student ID ____________ Name: ________________________

Please read the following instructions carefully.

- You have **90** minutes to complete this exam. This question booklet contains **6** questions for a total of 120 points. Check to see if any pages are missing.
- **Please write all answers in the answer book.**
- Read the instructions for individual questions carefully before answering the questions.

Question	Points	Score
1	10	
2	20	
3	20	
4	20	
5	20	
6	30	
Total	120	

1. (10 points) Solve the following linear program (LP) graphically. Clearly identify the feasible region, the optimal solution, and the optimal objective value..

$$
\begin{aligned}
\max_{x_1,x_2} \quad & x_1 + \frac{1}{2}x_2 \\
s.t. \quad & x_1 + 2x_2 \le 6 \\
& x_1 + 2x_2 \ge 0 \\
& 2x_1 - x_2 \le 4 \\
& 2x_1 - x_2 \ge -2
\end{aligned}
$$

2. (20 points) Consider the following primal LP:

$$
\begin{aligned}
\max \quad & 2x_1 + 3x_2 + x_3 + x_4 + x_5 \\
\text{s.t.} \quad & 2x_1 + x_2 + x_3 + x_4 \le 3, \\
& x_1 - x_2 + 2x_3 + x_5 = 7, \\
& x_1 + 2x_2 - x_3 + 3x_4 + x_5 \le 6, \\
& x_2 + x_3 - x_4 + 2x_5 = 7, \\
& x_1 \ge 0, \ x_2 \ge 0,
\end{aligned}
$$

where x_3, x_4, x_5 are unrestricted in sign. Suppose that y_1, y_2, y_3, and y_4 are the dual variables corresponding to the four constraints of the primal LP, respectively, and that the optimal dual solution is $(y_1^*, y_2^*, y_3^*, y_4^*) = (2, -1, 0, 1)$. Use the complementary slackness conditions to find an optimal primal solution.

3. (20 points) A relief agency needs to ship medicine boxes from two warehouses to three clinics. Warehouse 1 has 20 boxes available, and Warehouse 2 has 30 boxes available. Clinic 1 needs 10 boxes, Clinic 2 needs 25 boxes, and Clinic 3 needs 15 boxes. Since total supply equals total demand, all boxes must be shipped. The shipping cost per box is shown below.

	Clinic 1	Clinic 2	Clinic 3
Warehouse 1	4	6	8
Warehouse 2	5	4	3

Let x_{ij} denote the number of boxes shipped from Warehouse i to Clinic j.

(a) Formulate a linear program that minimizes the total shipping cost while satisfying all clinic demands.

(b) Write down the dual linear program. Let u_1, u_2 be the dual variables associated with the two warehouse constraints, and let v_1, v_2, v_3 be the dual variables associated with the three clinic constraints.

(c) Suppose the optimal shipment plan is $x_{11}^* = 10$, $x_{12}^* = 10$, $x_{13}^* = 0$, $x_{21}^* = 0$, $x_{22}^* = 15$, $x_{23}^* = 15$. Also suppose an optimal dual solution is $u_1^* = 2$, $u_2^* = 0$, $v_1^* = 2$, $v_2^* = 4$, $v_3^* = 3$. Explain the meaning of the dual variables intuitively. What is the economic interpretation of the dual LP and its optimal solution?

4. (20 points) A manufacturer produces a single product over the next 6 months. The demand in each month is known and must be satisfied on time. Shortages are not allowed. The manufacturer may produce in a month only if it sets up the production line in that month. If the production line is set up in month t, where $t \in \{1, 2, ..., 6\}$, the firm pays a fixed setup cost. In addition, it pays a variable production cost for each unit produced and an inventory holding cost for each unit carried from one month to the next.

   The monthly data are shown below.

Month	1	2	3	4	5	6
Demand	40	60	35	70	55	45
Production capacity	80	80	80	80	80	80
Variable production cost	12	11	13	10	12	14
Setup cost	300	300	300	300	300	300
Holding cost	2	2	2	2	2	–

   Assume that the initial inventory is zero and that the company wants no inventory left over at the end of Month 6. Formulate this problem as a mixed-integer linear program that minimizes the total production, setup, and inventory holding cost. Clearly define all decision variables.

5. (20 points) Consider a two-player zero-sum game where Player A's payoffs are shown below. Player A chooses rows ($A1$ to $A5$), and Player B chooses columns ($B1$ to $B5$).

Player A\Player B	B1	B2	B3	B4	B5
A1	4	1	5	2	6
A2	2	3	4	4	5
A3	1	0	2	1	3
A4	1	2	3	3	4
A5	0	0	1	1	2

   Answer the following questions:

   (a) Eliminate all dominated strategies for Player A and Player B.

   (b) Does the reduced game have a pure strategy Nash equilibrium? If yes, find the equilibrium; if no, explain why every payoff in the reduced game is unstable.

6. (30 points) A firm has options to invest in 10 different projects. The total investment cannot exceed the available budget q. The profits and costs of these projects are given by $(p_1, p_2, \ldots, p_{10})$ and $(c_1, c_2, \ldots, c_{10})$, respectively. The firm wants to maximize its total profit. In addition to the budget constraint, the managers need to consider the following requirements:

   (a) Projects 2 and 5 cannot be chosen together.

   (b) Exactly 4 projects are to be selected.

(c) At least one project must be selected from the project set $\{1, 3, 4\}$.

(d) If Project 6 is selected, then Project 7 must also be selected.

(e) If Project 8 is selected, then at least one of Projects 3 and 4 must also be selected.

(f) Either both Projects 1 and 10 are selected, or neither of them is selected.

(g) If both Projects 3 and 4 are selected, then Project 9 must also be selected.

(h) Project 5 cannot be selected unless at least two projects from the set $\{6, 7, 8\}$ are selected.

(i) At least one of the following two conditions must hold: at least two projects from the set $\{1, 2, 3, 4\}$ are selected, or at most one project from the set $\{5, 6, 7, 8\}$ is selected.

(j) If Project 10 is selected, then at least one of the following two conditions must hold:
   - Both Projects 2 and 9 are selected.
   - Neither Project 3 nor Project 4 is selected.

Define integer decision variables and formulate logical constraints for (a)–(j).